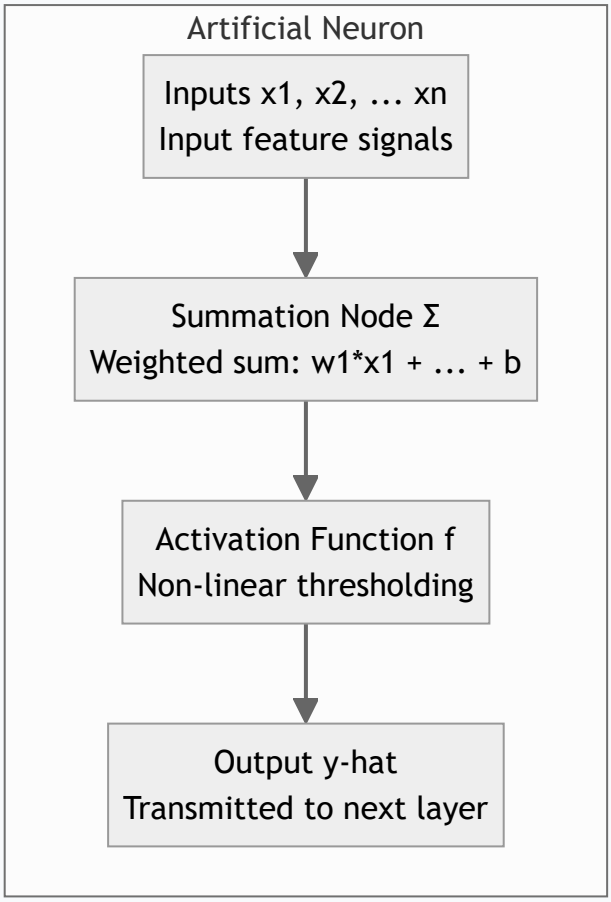
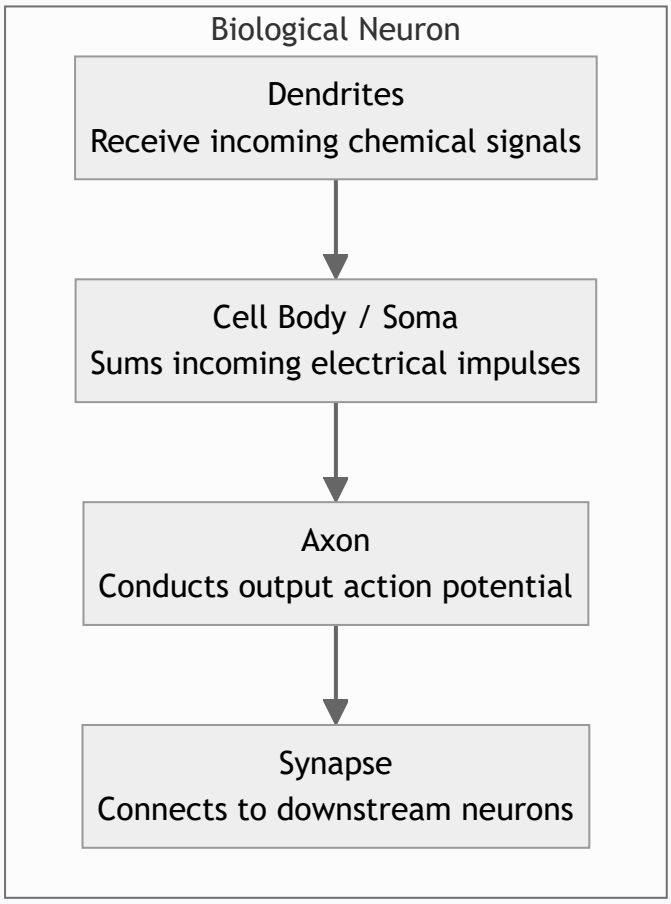
Submodule 36.03: What are Neural Networks?

1. Biological Inspiration: The Human Brain

The foundational architecture of Deep Learning is inspired by the human neurological system. The human brain contains approximately **86 to 100 billion biological neurons**, each interconnected through thousands of synapses, forming a massively parallel information processing network.

Biological vs Artificial Neuron Mapping:


Artificial Neuron
Inputs x1, x2, ... xn
Input feature signals
Summation Node Σ
Weighted sum: w1*x1 + ... + b
Activation Function f
Non-linear thresholding
Output y-hat
Transmitted to next layer

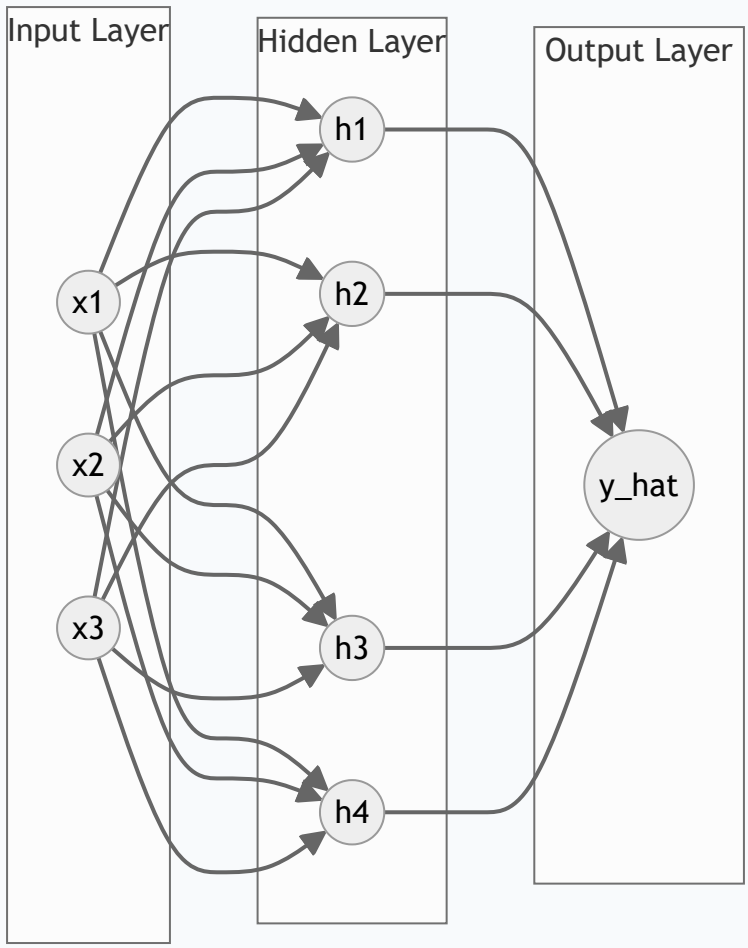

Biological Neuron
Dendrites
Receive incoming chemical signals
Cell Body / Soma
Sums incoming electrical impulses
Axon
Conducts output action potential
Synapse
Connects to downstream neurons

Biological Component	Artificial Neural Network Equivalent	Computational Role
Dendrites	**Input Features ($x_1, x_2, \ldots, x_n$)**	Receives data inputs from preceding nodes
Synaptic Weights	**Weights ($w_1, w_2, \ldots, w_n$)**	Controls the amplification/attenuation strength of each connection
Cell Body (Soma)	**Summation Node ($\sum w_i x_i + b$)**	Accumulates all weighted inputs plus bias offset
Action Potential Threshold	**Activation Function ($f(z)$)**	Decides whether and with what magnitude the neuron should fire
Axon	**Neuron Output ($\hat{y}$)**	Transmits computed activation signal to downstream layers

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2. Artificial Neural Network (ANN) Architecture

An **Artificial Neural Network (ANN)** is organized into discrete sequential groups of artificial neurons called **Layers**:



The Three Fundamental Layer Types:

1. **Input Layer**:
   - The entry point of the network.
   - Contains exactly one node for every input feature (e.g., if predicting house prices using 3 features: `square_feet`, `bedrooms`, `age`, the input layer has 3 nodes).
   - Performs no mathematical transformations; simply passes values forward.
2. **Hidden Layer(s)**:
   - Intermediate layers situated between input and output layers.
   - Perform mathematical dot-products, bias addition, and non-linear activation functions.
   - Extract increasingly abstract feature representations.
   - Networks with multiple hidden layers are called **Deep Neural Networks**.
3. **Output Layer**:
   - The final layer responsible for producing the model's prediction.
   - Node count and activation depend on the problem type:
     - *Binary Classification*: 1 node with **Sigmoid** activation (probability $\in [0, 1]$).
     - *Multi-Class Classification*: K nodes with **Softmax** activation (probability distribution summing to 1).

- *Regression*: 1 node with **Linear** (no) activation (continuous real number).